# Bishal Dhungana

West Palm Beach, FL | beesal13dh@gmail.com | linkedin.com/in/dbishal | dbishal13.github.io | github.com/DBishal13

## SUMMARY

Data Engineer with a geospatial specialization, building production data pipelines, cloud infrastructure, and spatial databases at utility scale. I design and maintain ETL/ELT pipelines, infrastructure-as-code frameworks, and data integration platforms handling millions of records with automated quality validation. Comfortable across the full data platform: ingestion, spatial/relational database design, API delivery, and cloud deployment.

## SKILLS

**Data Engineering:** Python, SQL, ETL/ELT Pipelines, PySpark, DuckDB, Databricks, Data Lakes, Parquet

**Geospatial:** ArcGIS Pro / Enterprise / Online, Utility Network, PostGIS, DuckDB Spatial, GDAL, PDAL, GeoPandas, Shapely, PyProj, LiDAR / Point Cloud, Global Mapper, LAStools, Overture Maps, Cloud-Optimized GeoTIFF, Folium, FME

**Cloud & DevOps:** AWS Lambda, S3, SageMaker, ECR / ECS, AWS CDK, Docker, IAM, VPC / Network Security, CI/CD

**Generative AI & Agentic Systems:** LangChain, LangGraph, RAG, Model Context Protocol, FastMCP, Vector Databases, Ollama, Chroma, Hugging Face Transformers

**Software Engineering:** JavaScript / TypeScript, R, FastAPI, React, Streamlit, Power BI

## EXPERIENCE

**Data Scientist (Geospatial & AI)** Apr 2025 – Aug 2026

*Florida Power & Light - Contract, West Palm Beach, FL*

- Built an API-based metadata pipeline with spatial (10m/30ft) and temporal (30-day) joins matching 350K+ condition reports to an 8M-image catalog, enriching ~2M images with condition metadata.
- Led the pipeline's move from ad hoc Python/GeoJSON scripts to an FME-to-PostGIS pipeline structuring 8M+ images and 65M+ associated annotations, with automatic time/version tracking via FME triggers. Designed the schema and cut 100+ metadata columns to ~50 with spatial indexing.
- Designed the annotation-storage schema for multiple model versions, and set up the taxonomy as a feature service with editor logging and concurrent multi-user updates, plus a separate read-only REST API for internal teams and vendors, replacing a fragile Excel-based process.
- Built a Streamlit app (later migrated to Power BI) with REST APIs summarizing dataset availability and taxonomy patterns for leadership.

**Associate Geospatial Analyst** Sep 2024 – Mar 2025

*Florida Power & Light - Contract, Jupiter, FL*

- Led QA/QC of LiDAR-derived power grid data for infrastructure resilience work, including pole hardening.
- Managed PostgreSQL/PostGIS databases, optimized Cloud-Optimized GeoTIFFs, and automated QA workflows in FME, ArcPy, and Model Builder to streamline data storage and validation.
- Built a FastAPI sidecar for heavier geoprocessing and format conversion.

**Research Assistant** Jan 2023 – Aug 2024

*Florida Atlantic University, Boca Raton, FL*

- Built an automated ArcGIS Pro geoprocessing pipeline, cutting processing time by 50%.
- Built a Python geocoding tool processing 3 million addresses, replacing a paid service; cut turnaround from 3 months to 1 week and saved $20K in licensing.

**Geomatics Engineer** Dec 2020 – Dec 2022

*Freelance Consultant, Kathmandu, Nepal*

- Designed municipal geodatabases (land use, soil, cadastral, resource maps) with topology and attribute rules.

## EDUCATION

**MSc, Geosciences — GIS Concentration** — Florida Atlantic University · 2023–2024

**BSc, Geomatics Engineering** — Tribhuvan University, Nepal · 2016–2021

## CERTIFICATIONS & TRAINING

- The Rise of the AI Data Engineer Bootcamp (2026)
- FAA Part 107 — Remote Pilot Certificate
- Machine Learning Specialization — DeepLearning.AI
- Deep Learning Specialization — DeepLearning.AI
- Remote Sensing Image Acquisition & Analysis — UNSW Canberra
- GIS Mapping & Spatial Analysis — U. Toronto
- Data Science with R — Johns Hopkins
- GIS Specialization — UC Davis
- Introduction to Databases — Meta

- **NASA ARSET Certifications (12):** Applied remote sensing training (2022-2024) in climate, disaster, and environmental risk modeling.
- **Esri Training Certifications (8):** GIS fundamentals, cartography, spatial data science, and imagery analysis.

## PUBLICATIONS

Dhungana, B.; Liu, W. *Urban–Rural Exposure to Flood Hazard and Social Vulnerability in the Conterminous United States*. ISPRS Int. J. Geo-Inf. 2024, 13, 339.

Mandal, A.K.; Thapa, A.K.; Thapa Chhetri, M.; Dhungana, B.; Bloetscher, F.; Yong, Y.; Su, H. *PyWMP: A Unified Python Framework for Multi-Fidelity 1D, 2D, and Hybrid Event-Based Flood Modeling*. Manuscript in review, DOI pending; ongoing contributor to the open-source PyWMP package.

## PROJECTS

**SurgeExposure:** A cloud-native pipeline overlaying NOAA storm-surge and flood-inundation data against Overture Maps building footprints, scoring per-asset flood exposure across 8 coastal regions — with a map and API. *(Python, DuckDB Spatial, FastAPI, PostGIS, AWS)* Demo | Repo

**geomlint:** A CI-first data-quality linter for vector geospatial files — catches invalid geometry, CRS confusion, and topology errors that mainstream data-observability tools skip. No GDAL required; pip install and it runs anywhere pip does. *(Python, Shapely, PyProj, PyPI, CLI)* Docs | Repo

**GTFS Transit Platform:** Data-engineering and analytics platform for GTFS transit feeds — headways, stop coverage, and spacing anomalies, explored in an interactive map app. Demo | Repo

**HazardMCP:** An MCP server unifying FEMA, USGS, NOAA, NWS, and NASA hazard data behind a single AI-agent interface — one-call location risk summaries and map rendering, compatible with Claude Desktop. *(Python, FastMCP, Folium, Docker)* Repo

**Road-Centerline:** A published Python package that extracts road centerlines from polygon geometries — worldwide, in any input CRS, in any format GeoPandas can read. CLI and Python API, computed via medial-axis skeletonization. *(Python, GeoPandas, pygeoops, PyProj, PyPI)* Repo

**GeoCoder:** Converts a spreadsheet of US street addresses into geocoded points and building footprint polygons using free Nominatim and Overpass APIs — as a no-install web app or a scriptable Jupyter notebook. *(GeoPandas, Nominatim / Overpass, Static Web App)* Demo | Repo